Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/03/2006
PI Name:	Sture, Stein Ph.D.		
Project Title:	Mechanics of Granular Materials		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Gravitational		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	stein.sture@colorado.edu	**Fax:**	FY 303-492-7317
PI Organization Type:	UNIVERSITY	**Phone:**	303-492-7651
Organization Name:	University of Colorado		
PI Address 1:	Dept. of Civil, Environ., and Architectural Engineering		
PI Address 2:	Engineering Center OT6-26, Campus Box 428		
PI Web Page:	http://bechtel.colorado.edu/		
City:	Boulder	**State:**	CO
Zip Code:	80309-0428	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	12/01/1992	**End Date:**	10/01/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	16
No. of PhD Candidates:	2	**No. of Master' Degrees:**	13
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Alshibli , Khalid	**Contact Phone:**	(225) 578-9179
Contact Email:	Alshibli@LSU.edu		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	4		
COI Name (Institution):	Al Shibli, K. (Louisiana State University)		
Grant/Contract No.:	NAS8-38779		
Performance Goal No.:			
Performance Goal Text:			
	Ground-based displacement-controlled triaxial experiments were conducted on a cohesionless granular material at the lowest effective confining pressures possible, that did not result in material instability, to assess constitutive properties, stability phenomena, and control parameters that will be applied to in-space experiments on 75 mm (diameter) and 150 mm (length) right cylindrical specimens. The ground-based tests on similar-sized specimens were conducted in the range 3.5-69 kPa, while the microgravity tests will be conducted at effective confinement levels in the range 0.05-1.30 kPa. The displacement-controlled mode of loading confined specimens was chosen mainly to maintain overall specimen-apparatus stability while strain-softening resulting from continuous or discontinuous bifurcation and discontinuous deformation fields are allowed to take place. Optical and other noncontacting displacement-sensing techniques are used to measure specimen response during experimentation. Prescribed displacements are transmitted in terms of loading, unloading, and reloading historics, while volume change is measured in "drained" tests, and pore fluid pressure is measured in "undrained" isochoric tests. Confinement pressure is transmitted to the granular material		

Task Description:	assembly through a thin flexible latex membrane surrounding the specimen. A subangular and uniform Ottawa quartz sand constitutes the specimen. Specimens tested both in space and on ground have been subjected to nondestructive and destructive (thin-slicing) testing to assess degrees of material uniformity and isotropy before and after experimentation. It appears that instability phenomenon associated with specimens of certain configurations result in curved internal surfaces of localized deformation and high rates of dilatancy, whose structure depends on bifurcation mode. Results from in-space experiments have shown that the friction angles at very low stresses can be as high as 75 degrees (range 75 to 56 degrees, with decreasing angles at higher stresses), while typical friction angles observed in terrestrial experiments at higher confinement levels are in the range of 42 to 40 degrees. Also, it was observed that the dilatancy angle, which is a measure of volumetric change and expansion was as high as 30 degrees, while typical terrestrial experiments show dilatancy angles in the range of 10 degrees. It was also observed that the elastic moduli were substantially higher (20 MPa) than was predicted for granular materials at very low confinement stress levels. It has also been observed that loose to medium dense granular assemblies tend to dilate rather than contract in volume at very low stress levels. Computed Tomography (Gamma-ray and X-ray) have shown extensive interior localized deformation or slip patterns in the specimens, which have not before been observed. The slip patterns, which clearly show local areas of localized volume expansion (increased porosity) appear in a very regular and periodic manner, and display turbine-fan like configurations with periodic spacing, which appear to be function of grain size, specimen shape, effective confining pressure and various moduli and material constants (constitutive properties). The curious observation is that these patterns become visible long after global material instability has occurred, and at the stage when the overall deformations have almost reached the residual strength or critical-state level. These findings are very different from what has earlier been observed in plane strain experiments, which show single interior surfaces, and coinciding instability and appearance of localized deformations. These findings have great importance in engineering mechanics and engineering disciplines, especially earthquake engineering. The objective of this research was to gain a quantitative and qualitative understanding of the mechanical behavior of cohesionless granular materials, which were either dry or water saturated, under very low effective confining pressures subjected to either drained or undrained conditions under static as well as cyclic conditions. The objective was also to understand how the very high friction and dilatancy angles arise, what the underlying mechanisms are, and how volume-change phenomena which are both counter-intuitive and in contrast to thermodynamics concepts can occur. The aim was also to understand how the water-saturated granular structure with relatively high strength, stability and stiffnes stransitions to a liquid mixture or composite with near zero strength and shear stiffness. The research has shown that the quasi-static deformation and strength behavior of granular materials at low stress levels is very different from conventional concepts and what current models predict. The findings will have impact on structure-medium analysis techniques and foundation design, especially in earthquake prone areas as well as broad areas of geophysics. The findings are also important in understanding phenomena related to strength and stiffness investigations of lunar and planetary soils (regolith).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The findings will help understand strength, stability and stiffness behavior of both dry and water-saturated granular soils subjected to static as well as dynamic (earthquake induced strong-motion) loading and deformations at low effective stress levels, which may influence the way foundations, earth structures, soil-structure systems, etc. are designed and constructed to enhance safety and economic designs.
Task Progress:	Theoretical and analytical research related to understanding basic phenomena related to the observed very high friction and dilatancy angles is continuing. Several models have bneen devised and are currently tested on data obtained on in-space experiments as well as terrestrially based experiments. We are also refining the calculations related to volume and configuration (geometrical) change data to gain a better understanding of dilatancy principles and phenomena as well as material instability modes.
Bibliography Type:	Description: (Last Updated: 03/14/2017)
Abstracts for Journals and Proceedings	Jeremic B, Yang ZH, Sture S. "Numerical Assessment of the Influence of End Conditions on Constitutive Behavior of Geomaterials." Am Society of Civil Engineers annual meeting, 2004. ASCE, Journal of Engineering Mechanics. 2004 Jun;130(6):741-745. , Jun-2004
Articles in Peer-reviewed Journals	Batiste SN, Alshibli KA, Sture S, Lankton M. "Shear band characterization of triaxial sand specimens using computed tomorgraphy." ASTM Geotechnical Testing Journal. 2004 Nov;27(6):568-79. , Nov-2004